AI531C — Week 11 Pre-Class Activity (PCM-11: The Restoration Bracket)

Prepares: Lecture 31 (Learned deblurring, Tue 27 Oct) · Lectures 32–33 (Super-resolution; Low-light), Wed 28 Oct · Tutorial T11 (Tue 27 Oct, training studio).

1 · Topic

The applied restoration week: deblurring, super-resolution and low-light — three tasks, one recipe, different forward models. Plus your degradation card for the T11 studio.

2 · Why you are preparing this

This week each task gets one hour. The shared recipe (pairs → model → loss → evaluate) is assumed from Lectures 27–30; preparing it lets each lecture go straight to what is SPECIAL about its task — the PSF, the invented pixels, the Poisson dark.

3 · Read (12–15 min)

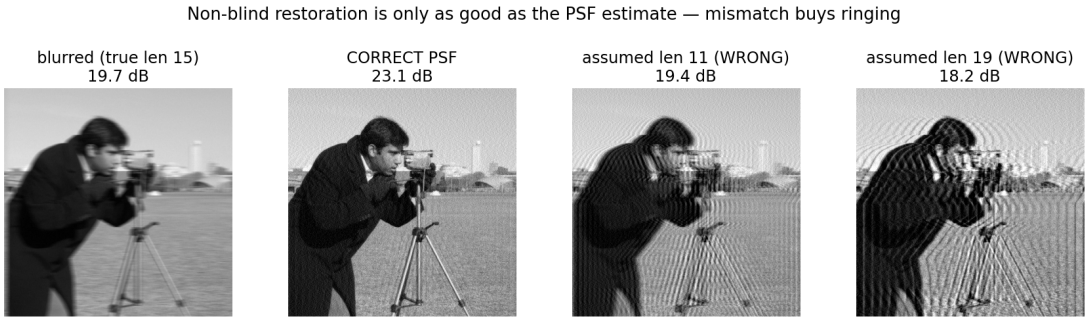
From the Comprehensive Self-Contained Lecture Notes:

– Lecture 31 chapter, the section on PSF mismatch (2 pages);

– Lecture 32 chapter, ONLY the forward-model page (LR = (HR∗k)↓s + η).

Optional depth: revisit Lecture 12's Wiener page — Tuesday's baseline.

4 · Interpret one figure (4 min)



Source: course figure, Lecture 31 (Wiener restoration under PSF mismatch).

In two sentences: compare the correct-PSF panel with the two mismatched ones — what artefact appears under mismatch, and which mismatch (too short / too long) looks worse here?

5 · One small calculation (5 min)

At ×2 super-resolution, what fraction of the output pixels were actually observed? At ×4? One line each.

6 · Predict (4 min)

Predict: can a Wiener-restored image ever score WORSE than the blurred input it started from? Commit (yes/no + one line) — Tuesday opens with the measured answer.

7 · Inspect this fragment (3 min)

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for L in (11, 15, 19): # assumed PSF length
print(L, psnr(clean, wiener(g, psf(L))))
# true length is 15
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Question: before running it — which of the three printed PSNRs must be highest, and is the ORDER of the other two obvious? Bring your guess.

8 · Bring one question to class

One question about your lane's restoration task — and your draft degradation card (the T11 studio trains from it that afternoon).

Submit: items 4–6 plus the draft card — one page maximum, before Tue 27 Oct, 9:00 AM.

Expected time: 30–35 minutes. Counts toward the weekly preparation portfolio (best 10 of PCM-2…PCM-14, completion-based micro-credit per the lesson plan). The short per-lecture primers in the course folder are optional warm-ups; only THIS card is submitted.